SOLace: How Topic Exposure Affects Downstream Testing Behavior in Standardized Test Practice

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***Abstract*—When a student encounters a difficult question, will their struggling end after that one question, or will the effects carry forward as they continue? A single difficult question can change students' testing behavior for multiple subsequent questions, but most analyses of practice test data treat the behavioral indicators on individual questions as independent of the surrounding questions. In this paper, we examine SOLace, a testing platform we designed for students in the Commonwealth of Virginia to study for their Standards of Learning (SOL) tests, and present our analysis of data collected from users throughout the 2026 SOL testing cycle. Through text embedding with Qwen3-Embedding-8B and the use of Markov chains, we reveal that encounters with questions of specific topics during testing are linked to behavioral changes that persist multiple questions later. These findings provide important information to educators on what topics tend to most strongly affect students' testing behavior.**

***Index Terms*—Data mining, educational technology, Markov processes, natural language processing.**

I. Introduction

Data mining techniques have long been applied to online educational environments to assess patterns in primary school students' interactions with online academic platforms. Oftentimes, researchers use these techniques to identify areas for improvement in the students' in-class learning or in the platform itself. However, few of these existing studies analyze how different content topics impact students' performance when questions are responded to sequentially. In this study, we investigate how text embedding and Markov chains can be used alongside one another to indicate the collective student body's strengths and weaknesses.

A. SOLace and SOL Testing in Virginia, United States

SOLace (learnsolace.org) [1] is a test preparation website we developed and released in 2025. SOLace is designed specifically to be used by students in the Commonwealth of Virginia for studying for Standards of Learning assessments, often referred to as "SOLs." These tests cover the content taught throughout the entire academic year and are typically administered in the spring a few weeks before schools begin summer break. The SOL assessments are most often administered during the school day and are not timed [2].

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SOLace offers a total of 40 practice tests that cover mathematics, English reading, and science content from grades 3 through 12. All practice tests come from fixed item sets and full assessments released by the Virginia Department of Education (VDOE) [3]. Tests are non-adaptive with fixed item orders, and students can only advance through questions; there is no option to return to previously viewed questions, which allows for a simpler sequential analysis [1]. Being an online website accessible from all devices, SOLace can be used in classrooms, by tutors, or by students at home. SOLace is a fully COPPA-compliant platform that does not collect personally-identifying student data such as IP addresses or session tokens.

B. Research Question

This paper aims to answer the broader questions of how students' behavioral interactions with practice tests in low-stakes testing environments (as evidenced by their accuracy, time spent per question, and their question revisions) connect to the topics of questions. We also explore how the knowledge of these behavioral patterns can assist teachers in providing proper support for students using practice test materials.

C. Code Availability

The data used in this study is available at https://github.com/iicebergg/solace-dataset. The code used to analyze this data is available at https://github.com/iicebergg/solace-research-code.

II. Methodology

The data for this study were collected from April 17th, 2026 through June 2nd, 2026. Table I shows the fields that were obtained from each of the 9,412 completed practice tests on the SOLace platform. Within each of these attempts, additional information listed in Table II was obtained for each of the 296,502 individual question responses within these tests. A total of 40 unique test forms are offered on SOLace.

A. Data Cleaning

Before any subsequent analyses were run on the test attempts themselves, we evaluated the test attempts to detect behavior that indicated a fully rushed attempt (i.e., no effort was put into the attempt whatsoever regardless of the student's position throughout the attempt) or a replay attempt (i.e., the

TABLE I
DATA COLLECTED ON INDIVIDUAL PRACTICE TESTS

Field	Variable Name	Meaning
Attempt Identifier	attempt_id	hexadecimal ID of attempt
Test Identifier	test_id	title of the test taken
Start Time	started_at	time of initialization
Completion Time	completed_at	time of submission
Correct Answers	score_correct	total correct answers
Total Questions	score_total	total questions the test contains

TABLE II
DATA COLLECTED ON INDIVIDUAL QUESTION RESPONSES

Field	Variable Name	Meaning
Question Number	question_number	the question's order on the test
Attempt Identifier	attempt_id	corresponds to test attempt ID
Correctness	was_correct	if the response was correct
Time Spent	time_seconds	seconds spent on the question
Answer Changes	answer_changes	times answer changed

student had already nearly completed the test in a different attempt beforehand and had memorized the questions). Replay attempts were classified as attempts that were in the top decile for speed and had an accuracy greater than or equal to 90%. Rushed attempts were classified as attempts that were in the top decile for speed and had an accuracy lower than 60%. Only responses from the cleaned attempts pool were used to construct the Markov chain.

TABLE III
CATEGORIZATION OF TEST ATTEMPTS FOR DATA CLEANING

Attempt Type	Attempts of Type	% of Uncleaned Attempts
Uncleaned attempts	9,412	100%
Rushed attempts	172	1.8%
Replay attempts	645	6.9%
Cleaned attempts	**8,595**	**91.3%**

B. Clustering of Questions by Topic

We used Qwen3-Embedding-8B to vectorize the questions' content and clustered the vectors with cosine K-Means into topics based on the original questions' semantic information. The elbow method on the within-cluster sum of squares together with cosine silhouette analysis was used to determine the optimal number of clusters (k) in each subject before using K-Means. The elbow method and cosine silhouette analysis revealed that the optimal number of clusters was 6 for Mathematics, 4 for Reading, and 11 for Science. Fig. 1 gives the Reading elbow graph as an example.

Table IV displays the clusters of words generated by K-Means for mathematics, reading, and science, respectively. We also generated 2D reductions of the clusters in the form of t-SNE plots; Fig. 2 shows the 2D math t-SNE plot as an example. All plots not pictured can be found in the repository. The **Top Keywords** column indicates the seven most semantically relevant words in the cluster using Class-based TF-IDF with SnowballStemmer. Because Qwen does not return topic keywords, c-TF-IDF helped to indicate many of

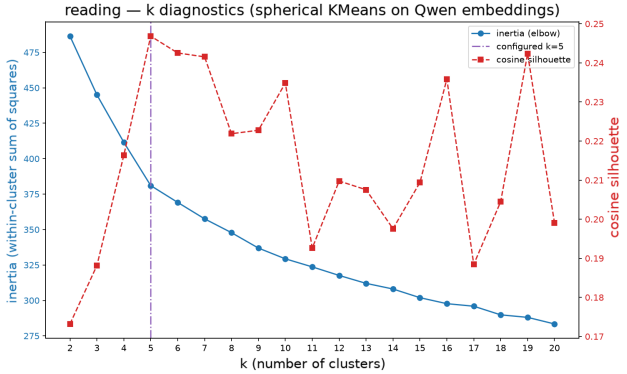


Fig. 1. Reading elbow plot

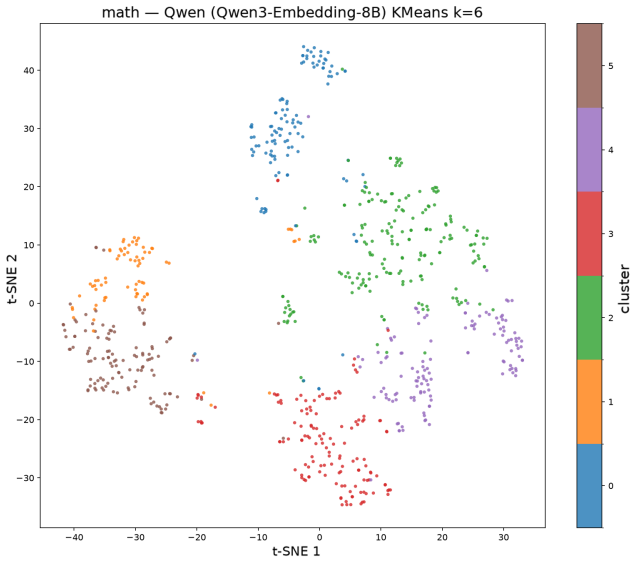


Fig. 2. Mathematics t-SNE plot. Each dot represents a question, and the cluster key shown on the side corresponds to the clusters described in Table IV.

the relevant words inside of each cluster, helping us to discern the **Topic**, which is the general idea of each cluster.

It is important to note that although some of the keywords may appear erroneous, the vast majority are fully topic-pertinent even if not immediately obvious. For example, the term "arrow" is shown to be significant to the Probability and Statistics cluster in Mathematics because Virginia often tests understanding of probability in elementary-aged students with questions about which portion of a spinner an arrow will land on.

C. Markov Chain

In order to categorize the behavior of students across each question, five observable states were defined. As the student submits their answer to a question, their state for that question response is recorded. To determine what state a student's submitted answer falls under, the question responses were categorized into states using the following order; that is, once the response is categorized, it will no longer be considered for the other states:

TABLE IV
QUESTION CLUSTERS BY SUBJECT WITH QWEN AND COSINE K-MEANS

Subject	Cluster	Size	Top Keywords	Topic
Mathematics	0	112	spinner, arrow, probability, labeled, data, mean, standard	Probability and Statistics
	1	73	prism, container, surface, rectangular, volume, cone, cylinder	3-Dimensional Geometry
	2	193	packages, difference, rule, fraction, number, including, tax	Word Problems
	3	137	function, relation, graph, domain, intercept, log, equation	Coordinate Plane
	4	128	step, solution, zero, equivalent, expression, solved, equation	Algebra
	5	119	angles, triangle, figure, parallel, angle, lines, congruent	2-Dimensional Geometry
Reading	0	158	according, useful, article, mountain, lake, hotel, report	Non-fiction
	1	87	poem, figurative, tone, speaker, example, language, lines	Poetry
	2	99	organized, author, article, paragraph, wrote, summarizes, purpose	Organization
	3	83	means, entry, root, synonym, dictionary, comes, read	Vocabulary
	4	114	benjamin, story, charlie, jared, character, leroy, theme	Fiction
Science	0	14	equator, latitude, hurricanes, location, pots, fishermen, north	Weather and Climates
	1	26	investigation, students, solutions, independent, probe, variable, faucets	Experimental Design
	2	35	cell, organism, dna, percentage, structures, heterozygous, trait	Cell Biology
	3	33	moles, formula, elements, equation, solution, reaction, element	Chemical Reactions
	4	21	ecosystem, zebras, tick, birds, tree, food, plant	Ecosystems
	5	15	tides, waves, earth, soccer, rays, visible, planets	Physics
	6	13	coal, action, reduce, ponds, stone, pollution, nonrenewable	Energy
	7	14	scientists, years, fossil, earth, organisms, happened, theorize	Paleontology
	8	15	formed, formations, rocks, mineral, trip, sinkhole, collecting	Geology
	9	20	oak, population, mass, wilt, density, speed, trees	Ecology
	10	16	wave, comb, objects, sled, toy, track, electromagnet	Mechanics

- **BF (blank fast):** multiple-choice questions that were left unanswered (answer_changes = 0) and were submitted in a time less than or equal to the rapid threshold. Indicates the student rapidly clicked through the question without providing an answer.
- **BS (blank slow):** multiple-choice questions that were left unanswered and were submitted in a time greater than the rapid threshold. Indicates the student may have deliberated over the question before deciding to leave it blank and move on.
- **R (rapid):** questions that were answered in a time less than or equal to the rapid threshold. Indicates rapid click-through behavior where despite providing a response, the student likely did not spend enough time to honestly consider the question.
- **V (reVision):** questions that had different answer choices selected greater than or equal to 3 times. Indicates potentially heavy deliberation and potential uncertainty through numerous answer changes before submission.
- **C (correct):** questions that meet none of the criteria for BF, BS, R, or V and were answered correctly. Indicates an honest and correct response from the student.
- **W (wrong):** questions that meet none of the criteria for BF, BS, R, or V and were answered incorrectly. Indicates an honest-but-incorrect response from the student.

We use a framework similar to Wise and Ma's normative threshold method to define rapid responses in the Markov chain [4]. Setting a fixed response time or percent of the average response time that indicates rushing can overlook the possibility of certain questions requiring more time than others to be mindfully responded to or the possibility that a different percent than the set percent of responses were rushed. In Wise and Ma's approach, the threshold for rapid guessing is instead defined as a certain percentage of the average response time of each individual question. This accounts for the natural variations in the complexity of each question while preventing a skewed fraction of responses from being flagged as rapid guesses. However, instead of using the mean response time as the center as Wise and Ma did, we use the *median* response time as the center to prevent extremely long response times (potentially caused by, for example, a response in which the user may have walked away from their device and come back later) from skewing the center. We defined 30% of the median with a cap of 10 seconds as the threshold for rushing.

The lifts for the next observable state that the student will occupy based on the state the student currently occupies are shown in Fig. 3. These lifts were calculated using the standard lift formula used in association rule mining [5]. If the lift is greater than 1, then there is a positive association between the cluster coinciding with a negative changepoint, and if the lift less than 1, then there is a negative association between the cluster coinciding with a negative changepoint. An answer submission in which the student transitions from one state to a different state will be referred to as a **changepoint** in this paper.

D. Relating Markov Chains to Qwen Clusters

In order to find the questions where students exhibited a negative change in their response behavior (i.e., a shift in observable state with a subsequent decrease in accuracy), we first identified all changepoints across all test responses where the accuracy dropped across the break, termed "negative changepoint responses." Then, we looked up the questions coinciding with each negative changepoint response and looked up its cluster. The per-cluster lift was identified by dividing

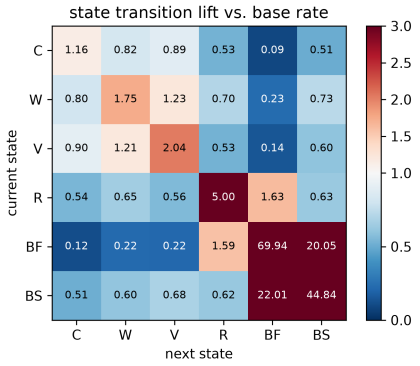


Fig. 3. Heatmap displaying the transition lifts between each observable state.

the rate of that cluster's questions coinciding with a negative changepoint by the rate of *all* of that subject's questions coinciding with a negative changepoint.

III. Results

By individual subject, the negative changepoints present across all responses in each cluster are shown in Table V. It is important to note that the topics span multiple test forms and grade levels. Therefore, the data are much less likely to be skewed by a question of a certain topic simply appearing before more difficult questions in a specific form.

Fig. 3 reveals a clear pattern in how students' approaches tend to change as they advance through the test. Generally, once a student begins submitting blank responses (regardless of whether they're answering quickly or slowly as in state BF or BS), they are *extremely* likely to continue to submit blank answers for the following questions. Similarly, if a student is submitting answers to questions but at a rapid pace (state R), they are also very likely to continue answering at a rapid pace, albeit less so than if they were answering blankly (lift of 5.00 versus a lift of at least 20.05). A student answering a question correctly at a reasonable speed (state C) says comparatively little about how the student will answer the next question; the student is only around 16% more likely than the average baseline to answer the next question correctly at a non-rushing speed.

Overall, students tended to exhibit lower response accuracy following changepoints; if they experienced a change in behavior, it was likely that their accuracy got worse instead of better. Fig. 4 indicates the frequency of changepoints within different positions throughout the tests. The position throughout the test was calculated as the fraction of the total test questions that the student had advanced to. As evidenced by this histogram, changepoints are generally less likely to occur in the bottom quintile of the tests' duration, but their frequency remains relatively consistent throughout the remainder of the test.

Fig. 5 displays the change, or delta, in response accuracy observed before and after the occurrence of changepoints. The histogram shows that students overall more frequently exhibit decreased accuracy following changepoints than increased accuracy following changepoints.

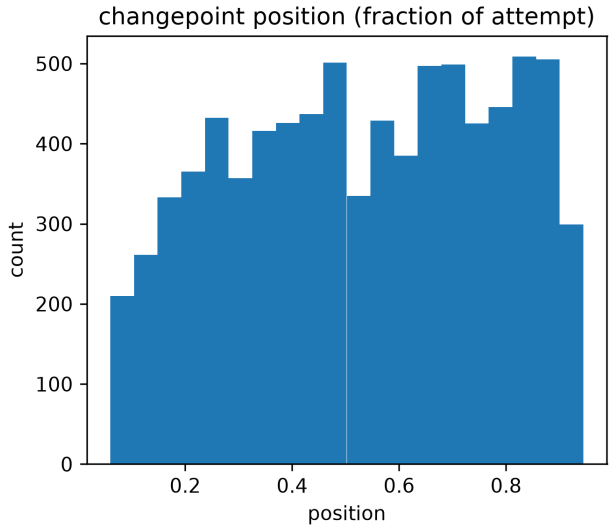


Fig. 4. Histogram displaying where changepoints tend to occur throughout the test

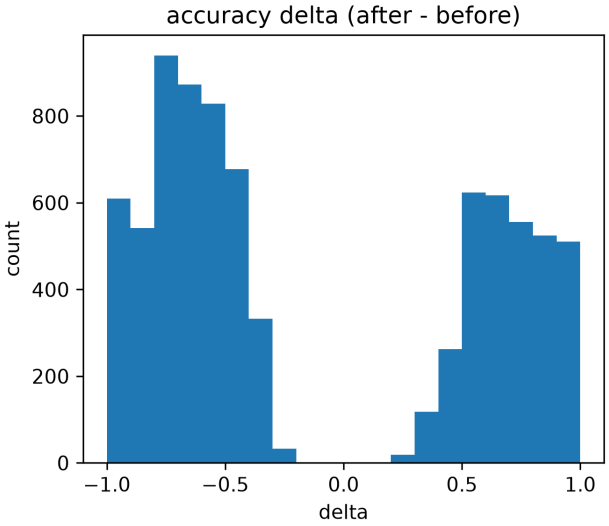


Fig. 5. Histogram displaying the distribution of change in response accuracy following changepoints

IV. Discussion

The results of this study indicate that exposure to questions of certain topics is associated with an increased probability for decreased accuracy going forward throughout the test (until the next changepoint occurs). Fig. 5 also indicates that when a change in answering behavior occurs mid-test, the responses following this change typically tend to be of lower accuracy than the responses prior to the change. The topics that were most positively associated with a decrease in accuracy (lift $\geq$ 1.25) were the Coordinate Plane and 2-Dimensional Geometry (clusters 3 and 5 in Mathematics) and Energy and Geology (clusters 6 and 8 in Science). The topics that were least associated with a decrease in accuracy (lift $<$ 0.75) were

TABLE V
NEGATIVE CHANGEPOINTS ACROSS TOPIC CLUSTERS, BY SUBJECT

Subject	Cluster	Topic	Total Responses	Neg. Changepoint Resp.	Lift
Mathematics	0	Probability and Statistics	24,570	422	0.9378
	1	3-Dimensional Geometry	13,373	298	1.2257
	2	Word Problems	54,465	761	0.7685
	3	Coordinate Plane	12,068	330	1.5041
	4	Algebra	9,868	163	0.9085
	5	2-Dimensional Geometry	14,568	373	1.4083
Reading	0	Non-fiction	22,443	493	1.1591
	1	Poetry	12,357	263	1.1230
	2	Organization	13,119	281	1.1302
	3	Vocabulary	11,092	155	0.7373
	4	Fiction	17,445	257	0.7773
Science	0	Weather and Climates	2,951	47	0.9949
	1	Experimental Design	6,640	131	1.2325
	2	Cell Biology	15,825	212	0.8369
	3	Chemical Reactions	2,695	53	1.2285
	4	Ecosystems	8,501	131	0.9626
	5	Physics	4,839	54	0.6971
	6	Energy	3,241	72	1.3878
	7	Paleontology	4,183	75	1.1201
	8	Geology	4,137	83	1.2533
	9	Ecology	5,886	71	0.7535
	10	Mechanics	7,132	128	1.1212

Vocabulary (cluster 3 in Reading) and Physics (cluster 5 in Science). We recommend teachers pay extra attention to these topics as they teach and consider how their students perform on tests in which they appear.

Our analysis was limited by the fact that students could not be tracked across test attempts. Laws such as the Children's Online Privacy Protection Act (COPPA) prevent persistent tracking of children under the age of 13, who comprise the primary user base of SOLace [6]. Performing an analysis with students being identifiable across test attempts (likely by COPPA's school consent exception) would strengthen the accuracy of the initial data cleaning by helping more clearly identify students who are replaying a test. Our categorization of responses into "rushed" categories may have also been skewed slightly because we only timed responses to the nearest second, not to the tenth or hundredth of a second. Timing responses more precisely may have allowed questions to be categorized with better accuracy.

Teachers in Virginia are prescribed curricula that provide them with the information they need to cover in their classes. However, these curricula often fail to identify which specific concepts students tend to struggle on most, leading to teachers needing to observe their classes' strengths and weaknesses to find out where to spend extra time. This leads to students struggling through their learning, which often cannot be remedied for every student before SOL tests are administered.

V. CONCLUSION

In this study, we collected data from student practice tests on SOLace and analyzed this data using LLM text embedding and Markov chains. Our findings indicate the collective weaknesses of students across the state of Virginia and which topics are associated with lower response accuracy not only for that individual question, but for multiple questions *after*. The finding that several specific concepts are positively associated with students struggling for not only those questions themselves, but for multiple questions *afterward* relative to the students' prior accuracy is worth expanding on in further research.

VI. ACKNOWLEDGMENTS

Much of the code used for the data analyses in this work was generated by Claude Code with the Opus 4.8 model. We reviewed all AI-generated code before running to confirm safety and modified code manually in many instances to adjust outputs. This paper is human-written with minor grammar checks done by Claude Opus 4.8. We take full responsibility for the final content of this work.

REFERENCES

[1] V. Berg, "SOLace," Learnsolace.org. Accessed: Aug. 12, 2026. [Online]. Available: https://learnsolace.org

[2] Virginia Assessment Program, "Spring and Summer 2026 Test Implementation Manual," Virginia.gov. Accessed: Aug. 12, 2026. [Online]. Available: https://www.doe.virginia.gov/home/showpublisheddocument/52269/639032106891270000

[3] "Released Tests & Item Sets (ALL SUBJECTS) — Virginia Department of Education," 2014. Accessed: Aug. 18, 2026. [Online]. Available: https://www.doe.virginia.gov/teaching-learning-assessment/student-assessment/sol-practice-items-all-subjects/released-tests-item-sets-all-subjects

[4] S. Wise and L. Ma, "Setting response time thresholds for a CAT item pool: The normative threshold method," presented at the Annu. Meeting Nat. Council Meas. Educ., Vancouver, BC, Canada, Apr. 2012.

[5] Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, and Vipin Kumar. *Introduction to Data Mining*. 2nd ed. Pearson, 2018.

[6] Federal Trade Commission, "Complying With COPPA: Frequently Asked Questions," Federal Trade Commission. Accessed: Sep. 02, 2026. [Online]. Available: https://www.ftc.gov/business-guidance/resources/complying-coppa-frequently-asked-questions